

Akai Force × FL Studio

Live Control script: User Manual



Version 1.2.0

github.com/Pyrodrifter/akai-force-fl-studio

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1. Overview

This script turns an **Akai Force** into a full controller for **FL Studio**, the same way the Force works with Ableton Live: its touchscreen shows your tracks, clips and plugin parameters, the 8×8 pads launch clips, sequence steps and play notes, and the knob screens show names and values.

How it works

The Force has a **Live Control** mode made for Ableton Live. In that mode it talks plain MIDI over the network through Akai's **Network MIDI driver**, on a port called `Akai Network - DAW Control`. This project re-implements the other side of that conversation as an FL Studio **MIDI controller script**, so FL Studio takes Ableton's place. Nothing else runs in the background: no server, no VST, no bridge app.



The Force draws its own screens. The script only sends names, colours and values, and reacts to your pads, knobs, buttons and touches.

What you get

- **Eight pad modes:** PERFORM (clip launching), PATTERNS, STEPS (step sequencer with per-step editing), KEYS (with scales), DRUMS, CHANNELS, PLUGIN (play with any plugin's parameters and presets) and SONG (jump to any bar).
- **Mixer on the Force:** 8 mixer inserts at a time with names, colours, volume, pan, mute, solo, arm, meters and up to 4 sends. Knobs control volume, and the knob screens show the name or the dB value.
- **Plugins from the Force:** 8 parameters at a time (macros first), preset browsing, open the plugin window, bypass, and randomize-with-undo, for the selected instrument or any mixer effect.
- **Transport:** play, stop, record, song record, overdub, loop record, count-in, tap tempo (it flashes on the beat), metronome, undo/redo, tempo entry and nudge, snap, song position, new pattern and markers.
- **Your own plugin pages:** choose and name the 8 knobs per plugin in `force_plugin_maps.py`.

2. Requirements

What	Details
Akai Force	With Live Control mode (the Ableton controller mode), connected to your network by Wi-Fi or a USB-to-Ethernet adapter.
Akai Network Driver	Free, from the inMusic Software Center (<i>My Hardware</i> tab) or akaipro.com. Available for Windows and Intel Macs only.
FL Studio	With Python MIDI scripting (FL Studio 20.7 or newer). Developed and tested on FL Studio 2026 (v26.1).
Computer	Windows 10/11 (tested). macOS on Intel should work but hasn't been tested. Apple Silicon Macs can't run Akai's network driver.
Python	Not needed for the script (FL Studio has its own). Only the optional developer tools use it.

One device at a time. If several Forces or MPCs are on the network, only one can be in Live Control with the computer at once. Close Ableton Live too, so it doesn't compete for the port.

3. Installation

Step 1: Akai Network Driver and pairing

1. Install **Akai Network Driver** (inMusic Software Center → My Hardware, or akaipro.com). Restart the computer.
2. On the Force, press **MENU**, tap the **gear** icon (Preferences), open **Wi-Fi** and join the same network as the computer. Or use a USB-to-Ethernet adapter and enable **Ethernet** in Preferences.
3. Open the **Akai Network Driver** app and select your Force (IP address and serial number) under **Configured Remote Device**. If it isn't listed, use **Add a Device** and type its IP address. On the Force, hold **SHIFT** and tap **Info** in the Wi-Fi menu to see it.

After this, Windows has two new MIDI ports: `Akai Network - DAW Control` (used by this script) and `Akai Network - MIDI`.

Step 2: Install the script

Pick whichever is easiest:

A. Release download (recommended)

1. Download **AkaiForceLive-v1.2.0.zip** from the *Releases* page: `github.com/Pyrodrifter/akai-force-fl-studio/releases`
2. Unzip it and double-click **install.bat**.

B. One line in PowerShell

```
irm https://raw.githubusercontent.com/Pyrodrifter/akai-force-fl-studio/main/install.ps1 | iex
```

C. macOS (Intel, untested)

```
curl -fsSL https://raw.githubusercontent.com/Pyrodrifter/akai-force-fl-studio/main/install.sh | sh
```

The installer copies the script files into FL Studio's user *Hardware* folder (it reads FL's own user-data location, normally `Documents\Image-Line\FL Studio\Settings\Hardware\Akai Force Live`), then checks that FL Studio and the Akai Network driver are present and prints the next steps.

Manual install: create the folder above and copy `device_AkaiForce.py`, `force_protocol.py` and `force_plugin_maps.py` into it. That's all the installer does. It only copies `force_plugin_maps.py` the first time, so your own plugin pages survive updates.

Step 3: Set up FL Studio

1. Open **Options** → **MIDI Settings**. If FL was already running, click **Update MIDI scripts**.
2. Under **Input**, select `Akai Network - DAW Control`, then set:

Setting	Value
Enable	On
Controller type	Akai Force (Live Control)
Port	Any free number, e.g. 1

3. Under **Output**, select `Akai Network - DAW Control` and give it the **same Port number**. This links the script's messages back to the Force; without it the Force stays on *waiting for communication*.
4. Leave `Akai Network - MIDI` alone; this script doesn't use it.

Step 4: Connect the Force

On the Force press `MENU` and tap **LIVE CONTROL**. Within a second or two the screen shows your mixer tracks and the knob screens flash **PERFORM**. You're connected.

Check it: in FL, **View** → **Script output**, tab *Akai Network - DAW Control*, should read `Akai Force script v1.2.0 loaded` followed by `Akai Force connected`.

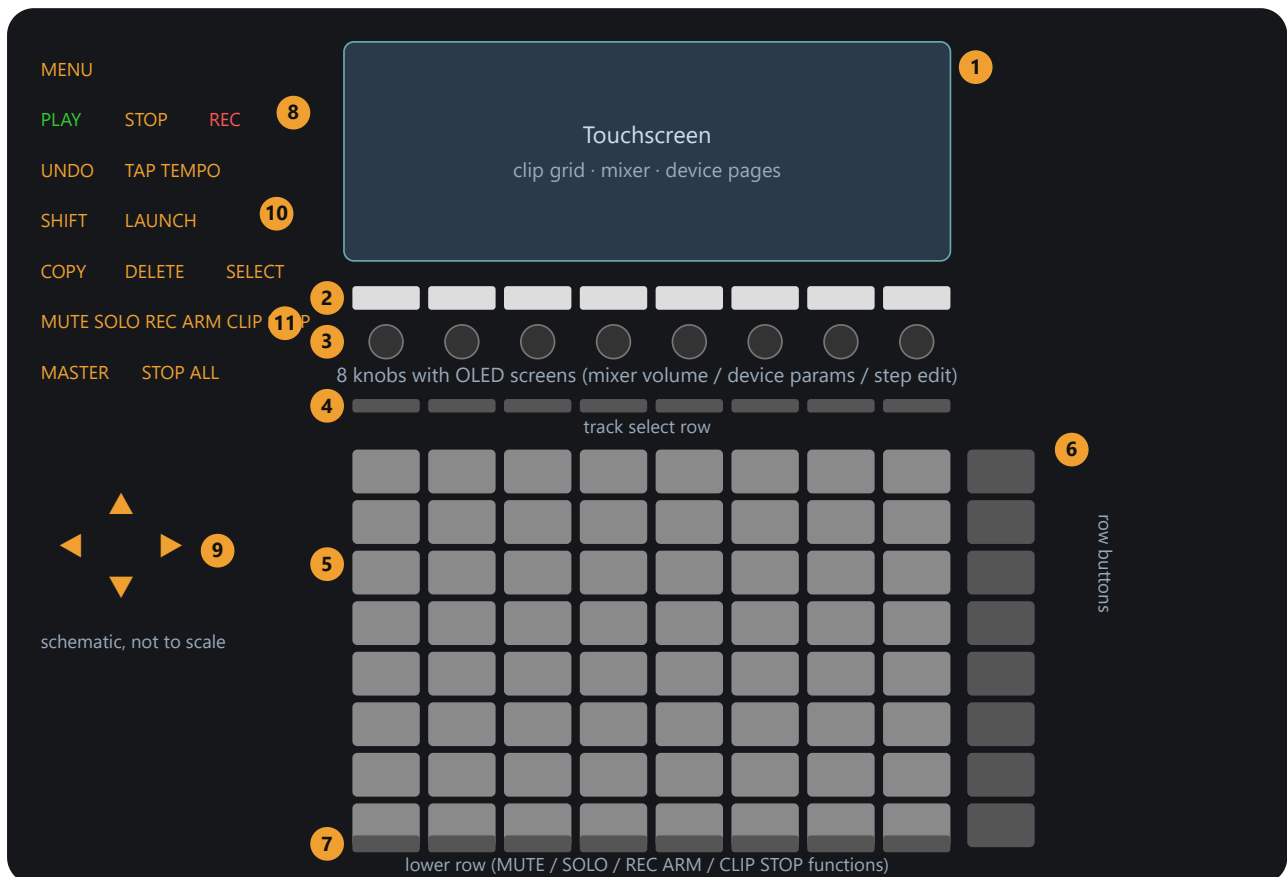
Updating and uninstalling

Update: run the installer again, then in FL click **Reload script** in Script output (or restart FL). **Uninstall:** `install.bat -Uninstall` (or `./install.sh --uninstall`), then set the `Akai Network - DAW Control` input back to (*generic controller*).

4. Your first session

New to this script? Start here. This chapter walks through the Force the way an FL Studio user thinks about it: what each part controls in FL, how to move around, and how to tell where you are.

A tour of the Force, in FL Studio terms



#	Part of the Force	What it does in FL Studio
1	Touchscreen	Shows FL's mixer tracks, the current pad grid with names, and the plugin page. Touch works: faders, buttons and grid cells all control FL.
2	Knob screens (OLEDs)	Tell you what each knob controls right now. They also flash the pad mode's name whenever you change modes (see below).
3	8 knobs	Mixer volume of 8 FL mixer tracks, or 8 parameters of a plugin, depending on the knob page.
4	Track select row	Selects that FL mixer track (like clicking it in FL's mixer). The selected one lights white.
5	8×8 pads	Depends on the pad mode : clips, patterns, steps, notes, drums, channels, plugin faders or song bars.
6	Row buttons (right of the pads)	An extra action for each pad row, different per mode (scales in KEYS, banks in PLUGIN, stop a track in PERFORM...).

7	Lower button row	Mute / solo / arm the FL mixer track above it (choose with MUTE, SOLO, REC ARM, CLIP STOP).
8	PLAY / STOP / REC	FL's transport, exactly like the buttons at the top of FL.
9	Arrow buttons	Move around: which 8 mixer tracks you see (◀▶) and the rows of the current mode (▲▼).
10	SHIFT / LAUNCH	LAUNCH changes the pad mode. SHIFT unlocks second functions (shown as SHIFT+... in this manual).
11	MUTE / SOLO / REC ARM / CLIP STOP, COPY, DELETE, SELECT, UNDO...	Editing helpers: see chapter 5.

Getting to each screen

The Force draws three main pages. Switch between them with the **icons along the left edge of the touchscreen**:

Icon	Page	What you see
Grid of dots (top)	Clip grid	The current pad mode on screen: every pad with its name and colour. Tap cells like pads.
Vertical bars (middle)	Mixer	8 FL mixer tracks: faders, pan, meters, mute/solo/arm and sends.
Rectangle (bottom)	Device	The current plugin: its name, preset, bank and 8 parameter faders.

The **knobs** have pages too: on the mixer page they set mixer volume, on the device page they set plugin parameters. Use the Force's **KNOBS** button (or its screen) to switch. The knob screens always show which you're on: track names mean mixer, parameter names mean plugin.

Which pad mode am I in?

Press **LAUNCH** to go to the next pad mode and **SHIFT** + **LAUNCH** to go back. The order is **PERFORM** → **PATTERNS** → **STEPS** → **KEYS** → **DRUMS** → **CHANNELS** → **PLUGIN** → **SONG**, then round again. Every time you change mode, **all 8 knob screens flash the mode's name for about a second**. After that, the labels in the right-hand column of the clip grid tell you where you are:

Mode	Knob screens flash	Right-hand column shows	Pads show
PERFORM	PERFORM	Playlist track names	Performance-Mode clips (or <i>Turn on Perf Mode</i>)
PATTERNS	PATTERNS	Pat 1-8, Pat 9-16...	One pad per pattern
STEPS	STEPS	Channel names	Steps of the current pattern
KEYS	KEYS	Scale names (Chromatic, Major...)	Note names (C3, D3...)
DRUMS	DRUMS	Note names	Four coloured 4×4 drum banks
CHANNELS	CHANNELS	Ch 1-8, Ch 9-16...	Channel names

PLUGIN	PLUGIN	Bank 1, Bank 2...	8 vertical faders with parameter names
SONG	SONG	Bars 1-8, Bars 9-16...	Bar numbers / marker names

What the knob screens tell you

When	Knob screens show
Normally (mixer knobs)	The name of the FL mixer track each knob controls, with a volume bar.
Touching a knob	That track's volume in dB (e.g. <i>-3.2 dB</i>), until you let go.
Device page	Plugin parameter names; the value while you touch the knob.
Changing pad mode	The new mode's name on all 8 screens (PERFORM, STEPS...).
Holding a step (STEPS)	That step's settings: Pitch, Vel, Release, Fine, Pan, Mod X, Mod Y.
Changing preset (PLUGIN)	The preset name, or NEXT PRESET if the plugin doesn't report names.
Other short messages	AT ON/OFF (pad pressure), BYPASS/ON (plugin), RANDOM/RESTORED, FX CHAIN/CHANNEL, PERF OFF, window names (ASSIGN A), SNAP..., tempo after a nudge, LOCKED/UNLOCKED, FOLLOW ON/OFF, COPIED, LOOP n (STEPS).

FL Studio's **hint bar** (top-left of FL's window) also says what just happened, for example *Force: row = solo* or *Force: randomized 8 parameters*.

Your first five minutes

1. **Mixer:** turn knob 1. FL mixer track 1's fader moves. Touch the knob to read the dB value. Press ► to see the next 8 tracks.
2. **Beat:** press **LAUNCH** until the screens say **STEPS**. Row 1 is your first channel (e.g. a kick): tap pads 1 and 5 to add steps, then press **PLAY**. The lower button row shows the playhead moving.
3. **Tweak a step:** hold one of those pads and turn knob 2 to change its velocity; knob 1 changes its pitch.
4. **Play notes:** **LAUNCH** to **KEYS**, pick a channel with a synth in FL (or with SHIFT+pad in CHANNELS), choose **Minor** with the second row button, and play. Every pad is now in key.
5. **Sound design:** **LAUNCH** to **PLUGIN**. Each column is a synth parameter: tap high or low to set it. ▼ tries the next preset; **SELECT** opens the synth's window in FL; SHIFT+DELETE randomizes, UNDO takes it back.
6. **Arrange:** **LAUNCH** to **SONG**, press **PLAY**, and tap any bar to jump there.

FL Studio → Force cheat sheet

I want to... (in FL)	On the Force
Play / stop / record	PLAY / STOP / REC
Undo / redo	UNDO / SHIFT+UNDO
Change the tempo	Tap TAP TEMPO, or type the value on the touchscreen
Turn the metronome on	SHIFT+TAP TEMPO

Show the mixer / channel rack / playlist / piano roll	ASSIGN A (steps through them; SHIFT goes back)
Switch pattern / song mode	ASSIGN B
Record into the song	Arrangement record on screen
Change the snap	Quantize value on screen
Make a new pattern / add a marker	Insert scene on screen / SHIFT + insert scene
Select a mixer track	Track select row (above the pads)
Move a mixer fader	Knob on the mixer page, or the fader on the touchscreen mixer
Mute / solo a mixer track	MUTE or SOLO, then the lower button under that track
Select a channel in the channel rack	CHANNELS mode: SHIFT+pad. STEPS: right-hand row button. Device page: prev/next device
Program steps	STEPS mode
Play the selected channel	KEYS or DRUMS mode
Switch pattern	PATTERNS mode (while playing, it switches at the next bar)
Clone a pattern	COPY
Quantize	SHIFT + first lower-row button, or Quantize on screen
Tweak a plugin	PLUGIN mode, or the knobs on the device page
Change a plugin's preset	PLUGIN mode ▲▼, or SHIFT + prev/next bank on the device page
Open a plugin's window	SELECT
Bypass a mixer effect	Device page on/off button (in FX-chain mode)
Launch clips live	PERFORM mode (turn on Performance Mode first)
Jump around the song	SONG mode, or the position control on screen

5. Controls

Hardware buttons

Control	Action
LAUNCH	Next pad mode (the mode name flashes on the knob screens). SHIFT+LAUNCH: previous mode.
PLAY / STOP / REC	FL transport. PLAY lights while playing.
TAP TEMPO	Tap tempo. SHIFT+TAP TEMPO: metronome on/off. Flashes on every beat while playing.
ASSIGN A	Show the next FL window: mixer → channel rack → playlist → piano roll (SHIFT: back). Change it with <code>ASSIGN_A</code> (see Settings).
ASSIGN B	Pattern / song mode (lit in song mode). Change it with <code>ASSIGN_B</code> .
UNDO	Undo. SHIFT+UNDO: redo.
◀▶	Move the 8-track mixer window by one insert (SHIFT: by 8). In PERFORM they scroll blocks and in STEPS they page the steps. There, SHIFT+◀▶ moves the mixer.
▲▼	Scroll the pad rows of the current mode (playlist tracks, pattern pages, channels, octaves, drum banks, song pages). SHIFT scrolls further, or by a semitone in KEYS.
Track select row	Selects that mixer insert in FL (the selected one is lit white).
MUTE / SOLO / REC ARM / CLIP STOP	Choose what the lower button row does for the 8 visible inserts. In PERFORM they also choose what the row buttons on the right do.
SHIFT + first lower button	Quantize the selected channel.
COPY	Clone the current pattern. SHIFT+COPY in PLUGIN: print the plugin's parameters for <code>force_plugin_maps.py</code> (chapter 8).
SHIFT+DELETE	In STEPS: clear the selected channel's steps in this pattern. In PLUGIN: randomize the 8 parameters.
SELECT	Open the current plugin's window in FL (press again to close an instrument's window).
SHIFT+SELECT	Pad pressure → aftertouch on/off (SELECT lights when on). For KEYS and DRUMS.
MASTER	Select the master track (lit while it's selected).
STOP ALL	PERFORM: stop all clips. Other modes: stop the transport.

Touchscreen

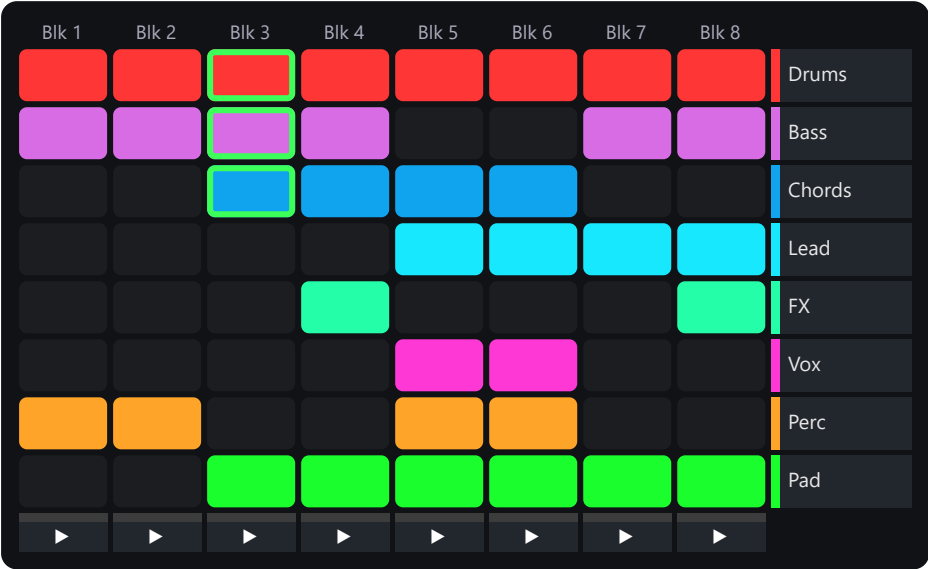
Use the icons on the left of the Force's screen to switch between its **clip grid**, **mixer** and **device** pages. The script fills them all:

Screen element	In FL Studio
Clip grid	Mirrors the pads of the current mode, with names. Tapping a cell = pressing that pad.
Track headers	Mixer insert names and colours.
Faders, pan, mute, solo, arm	The 8 visible mixer inserts.
Sends A-D	Send levels to your send/FX inserts (see Mixer).
Tempo	Shows FL's tempo; type a new value to change it.
Position encoder	Moves the song position a beat per tick (SHIFT: one step).
Metronome / loop buttons	Metronome on/off; loop button switches pattern/song mode (lit in song mode).
Quantize / Delete	Same as the hardware functions above.
Overdub	FL's overdub recording on/off. SHIFT: loop recording. (FL can't tell scripts whether overdub is on, so this button doesn't light.)
Automation arm	FL's count-in before recording, on/off (lit when on). FL has no script control of automation recording.
Arrangement record	Record into the song: switches FL to song mode, arms recording and starts playback. Again: stop recording.
Follow	The STEPS, PATTERNS and SONG pages follow the playhead (lit when on).
Device lock	Keep the device page and PLUGIN mode on the current plugin, whatever you select next (lit when locked).
Nudge – / +	Tempo down / up 1 BPM (SHIFT: 0.1 BPM).
Quantize value	Step FL's snap: none, 1/4 step, 1/2 step, step, 1/2 beat, beat, bar (SHIFT: back).
Insert scene	New empty pattern. SHIFT: add a playlist marker at the playhead.
Device page	8 plugin parameters with names and values, bank and device buttons.

6. Pad modes

The mode applies to the 8×8 pads, the clip grid on screen, and the row buttons to the right of the pads. The mixer, knobs and transport work the same in every mode.

PERFORM: clip launching

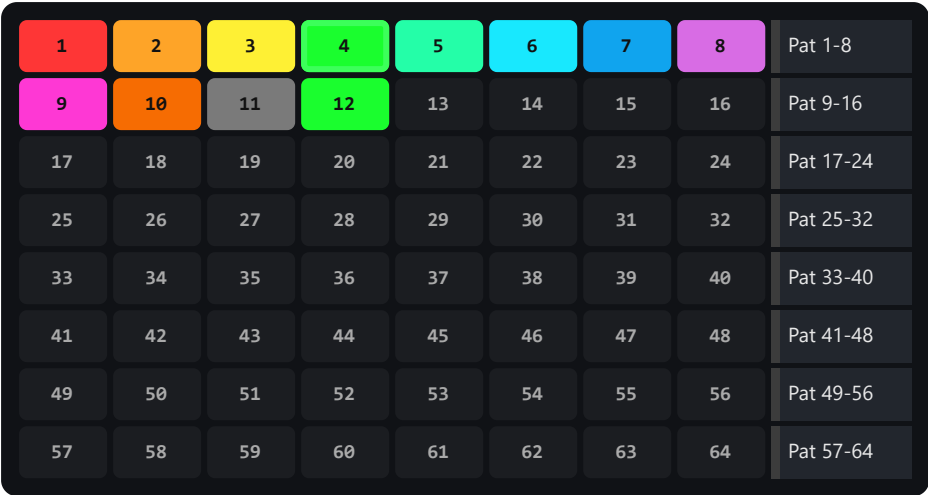


PERFORM: rows are playlist tracks, columns are Performance-Mode blocks. Pulsing = playing.

Control	Action
Pad	Launch that block's clip on that playlist track (other clips on the track stop). Release follows FL's trigger mode.
SHIFT + pad	Launch the whole column (every track's clip in that block), like an Ableton scene.
Row buttons (right)	Mute, solo or stop that playlist track, depending on the MUTE / SOLO / CLIP STOP button.
Lower row, CLIP STOP mode	Launch the block column above the button.
▲▼ / ◀▶	Scroll playlist tracks / blocks. FL outlines the visible area in the playlist.

PERFORM needs FL's **Performance Mode** switched on (Playlist → Performance mode). If it's off, the screen says *Turn on Perf Mode* and the pads do nothing.

PATTERNS



PATTERNS: one pad per pattern. The current pattern pulses.

Tap a pad to select that pattern (an empty slot creates it). ▲▼ pages through 8 patterns at a time (SHIFT: 64). While playing in pattern mode, a pad **queues** its pattern: it blinks, and FL switches to it at the start of the next bar, so you stay in time. Press it again or use SHIFT+pad to switch right away. Turn this off with `PATTERN_QUEUE = False`.

STEPS: step sequencer



STEPS: rows are channels, columns are steps. Lit = step on. The button row under the pads shows the playhead.

Control	Action
Tap a pad	Toggle that step. Lit = on, dark = off, exactly as in FL's channel rack.
Hold a pad + turn knobs	Edit that step: knob 1 pitch, 2 velocity, 3 release, 4 fine pitch, 5 pan, 6 Mod X, 7 Mod Y. The knob screens show the values while you hold. Editing never switches the step off.
Tap hard / soft	A new step takes the pad's velocity: hit harder for an accent (<code>STEP_PAD_VELOCITY</code>).

Hold a pad + tap another in the row	Fill every step between the two.
Hold a row button + press another	Copy the first channel's steps (with pitch, velocity etc.) to the second.
Hold a row button + tap a pad	Loop that channel after that step; the same step again removes the loop. Off by default: set <code>STEPS_CHANNEL_LOOP = True</code> (it uses an undocumented FL function).
SHIFT + pad / row button	Select that channel.
◀▶	Page through the pattern 8 steps at a time (up to its real length). With Follow on, the page follows the playhead while playing.
▲▼	Scroll channels (SHIFT: 8 at a time).
SHIFT+DELETE	Clear the selected channel's steps.

Channels whose notes were written in the **piano roll** are labelled (*piano roll*) and show no steps, just like FL's channel rack. The script treats a synth plugin whose notes use more than one pitch as piano roll. If that hides a real step pattern, set `STEPS_HIDE_PIANO_ROLL = False` (see Settings).

KEYS: melodic playing



KEYS (Chromatic): rows are a 4th apart, bottom-left is C3. Root notes in blue. The right-hand buttons choose the scale.

Plays the **selected channel**, and records like any MIDI keyboard. The pads are velocity sensitive.

Control	Action
Row buttons (right)	Pick the scale: Chromatic, Major, Minor, Dorian, Mixolydian, Harmonic minor, Major pentatonic, Minor pentatonic. In a scale, every pad is in key and rows are 3 scale degrees (a 4th) apart.
▲▼	Octave up / down. SHIFT+▲▼ moves the root by a semitone (the bottom-left pad is the root).
Colours	Root notes use the channel's colour (deep blue if it has none), naturals are grey and sharps darker.

DRUMS



DRUMS: four 4x4 banks of 16 notes. Bottom-left bank starts at C3 (MIDI 36), the standard FPC layout.

For FPC, Slicex and drum plugins on the selected channel. ▲▼ moves by a bank of 16 notes.

CHANNELS



CHANNELS: one pad per channel-rack channel. Tap to play, SHIFT+tap to select.

Each pad triggers one channel-rack channel. Perfect for kits built from separate samples. SHIFT+pad selects the channel (the selected one blinks). ▲▼ pages by 8.

PLUGIN: play with any plugin



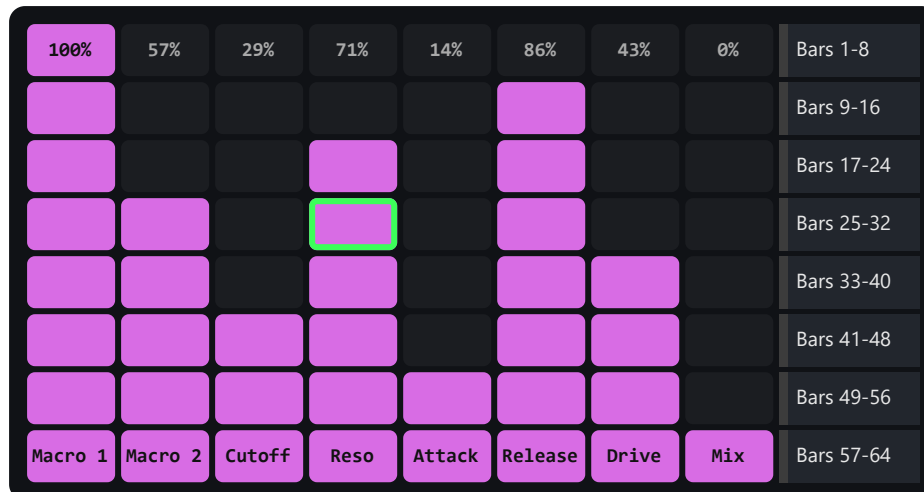
PLUGIN: each column is one parameter of the current bank; the lit height is its value. Names along the bottom, values along the top.

PLUGIN turns the pads into 8 faders for the plugin you're working on: the **selected channel's instrument**, or, in FX-chain mode, an **effect on the selected mixer track**. The most useful parameters come first: macros and master controls.

Control	Action
Pad	Set that column's parameter to that height (bottom = minimum, top = maximum).
Knobs (device page)	Fine control of the same 8 parameters.
Row buttons (right)	Choose parameter bank 1-8. SHIFT+▲▼ steps through banks beyond 8.
▲ ▼	Previous / next preset . The knob screens flash its name.
◀ ▶	Previous / next plugin (the next channel, or the next effect slot in FX-chain mode).
SELECT	Open the plugin's window in FL so you can see what you're changing.
SHIFT+DELETE	Randomize the 8 parameters on screen. Press UNDO straight after to put them back.

FL's own undo doesn't record plugin knob moves, so the script keeps a copy of the values before a randomize. Only the **last** randomize can be undone, and only until you touch another parameter.

SONG



SONG: one pad per bar. Marker sections take the marker's colour; the current bar pulses.

Each pad is one bar of the song. Tap to jump there. The current bar pulses and the page follows playback. Entering SONG switches FL to song mode. With **playlist markers**, each marker's bar shows its name and the section takes its colour. Without markers, 4-bar phrases alternate shades. ▲▼ pages by a row (SHIFT: 64 bars).

FL can't tell scripts where markers are, so the script finds them when you enter SONG mode (only while stopped), briefly moving the playhead and putting it back.

7. Mixer, knobs and screens

Feature	Details
Mixer window	8 consecutive mixer inserts, moved with ◀▶ (SHIFT: 8). FL outlines them in the mixer.
Knobs (mixer)	Volume of the 8 inserts. Hold SHIFT for fine control. The Force chooses whether knobs are on the mixer or device page (its KNOBS button / screen pages).
Knob screens	Show the insert name, the dB value while you touch the knob, and a volume bar.
Meters	Left/right peak meters for each strip on the mixer page.
Mute / solo / arm	On screen, or with the lower button row in MUTE / SOLO / REC ARM mode.
Sends A-D	Route each strip to up to 4 send inserts. By default these are the first inserts whose names contain <i>send</i> , <i>reverb</i> , <i>verb</i> , <i>delay</i> , <i>bus</i> or <i>fx</i> . Or list them in <code>SEND_TRACKS</code> . Turning a send on an unrouted strip creates the route.

8. Device page (plugins)

Control	Action
Knobs (device)	8 parameters of the current plugin. The knob screens show names, or values while touched.
Prev / next bank	Next 8 parameters (the bank number shows as <i>Bank 2/12</i> , plus the preset name when the plugin reports one).
SHIFT + prev/next bank	Previous / next preset.
Prev / next device	Select the previous/next channel, or the previous/next effect slot in FX-chain mode.
SHIFT + prev/next device	Switch between the selected channel's plugin and the selected mixer insert's effect chain (the knob screens flash FX CHAIN or CHANNEL).
Device on/off button	Bypass the effect, or mute the instrument's channel. Press again to turn it back on.
SELECT (hardware)	Open the plugin's window in FL.
Device lock (screen)	Stay on this plugin while you select other channels or inserts. Press again to unlock.
Focused effect window	If an effect's window is focused in FL, the page follows it automatically.

Only real parameters are listed: empty slots and FL's generic *MIDI CC / MIDI Channel* entries (thousands of them on VSTs) are skipped, and parameters named *Macro...* or *Master...* come first, so bank 1 of most synths starts with its macros.

Your own knob pages

For the plugins you use most, you can choose the 8 knobs yourself, in named pages, in `force_plugin_maps.py` (next to the script in FL's Hardware folder):

```
PLUGIN_MAPS = {
    "Sytrus": [
        ("Filter", ["Cutoff", "Resonance", "Env", "", "", "", "Drive", "Mix"]),
        ("Amp", ["Attack", "Decay", "Sustain", "Release"]),
    ],
}
```

1. Select the plugin, go to PLUGIN mode and press **SHIFT** + **COPY**. The script prints an entry with every parameter in **View** → **Script output**.
2. Paste it into `force_plugin_maps.py`, keep and reorder the parameters you want (up to 8 per page), and name the pages.
3. In FL, click **Reload script** in Script output.

Your pages come first, with their names on the screen and on the PLUGIN row buttons; the automatic banks follow (`APPEND_ALL_PARAMS`). Parameters are matched by name (not case-sensitive) or by number, and a name that isn't found is reported in Script output. The installer never overwrites this file.

9. Colours

Everything on the Force takes its colour from FL: channel colours (STEPS, CHANNELS, KEYS roots), mixer insert colours (track headers, track buttons), playlist block and track colours (PERFORM) and pattern colours (PATTERNS). The Force has a fixed 70-colour palette, so each FL colour is matched to the nearest one.

Give your channels real colours. FL's default channel grey can only become a pale grey-white on the pads. Bright, saturated colours (red, orange, cyan, green, purple...) look best. Colour each mixer insert like the channel that feeds it and a sound keeps its colour across every mode.

Dark colours are brightened automatically so pads never look unlit (`MIN_BRIGHTNESS`).

10. Settings reference

Advanced options live at the top of `device_AkaiForce.py` in FL's Hardware folder. Edit them in any text editor, then click **Reload script** in View → Script output. Re-running the installer overwrites them.

Setting	Default	What it does
PAD_AFTERTOUCH	False	Start with pad pressure → aftertouch on (also toggled live with SHIFT+SELECT).
PARAM_FIRST	macro, master	Parameters whose names start with these come first on the device page and in PLUGIN.
PARAM_SKIP	MIDI CC, MIDI Channel	Parameter names that are never shown.
SEND_TRACKS	()	Mixer inserts for send knobs A-D, e.g. (20, 21). Empty = find them by name.
SEND_KEYWORDS	send, reverb, ...	Words that mark an insert as a send when auto-detecting.
STEPS_HIDE_PIANO_ROLL	True	Hide piano-roll channels' notes in STEPS.
STEP_PAD_VELOCITY	True	New steps take the pad's velocity.
STEPS_CHANNEL_LOOP	False	Hold a STEPS row button + tap a pad to loop that channel (undocumented FL function).
ASSIGN_A / ASSIGN_B	windows / song_mode	What ASSIGN A and B do: <i>windows, song_mode, mixer, channel_rack, playlist, piano_roll, browser, metronome, loop_record, step_edit, none</i> .
FOLLOW	True	Start with Follow on.
PATTERN_QUEUE	True	PATTERNS pads switch at the next bar while playing.
NUDGE_BPM	1.0	Tempo change per nudge press.
MIN_BRIGHTNESS	200	Colours darker than this (0-255) are brightened.
SCALES	8 scales	The scales offered in KEYS (name + semitones).
VOLUME_STEP / PARAM_STEP	0.004 / 0.008	How far a knob tick moves volume / plugin parameters.
FINE	0.2	SHIFT multiplier for knobs.
FLASH_TIME	1.2 s	How long the mode name stays on the knob screens.
PING_INTERVAL / PONG_TIMEOUT	1 s / 8 s	Connection keep-alive and how long before it counts as lost.
HEAL_INTERVAL	0.25 s	Background refresh of one pad row, repairing any dropped network message.
FX_SLOTS / MAX_PARAM_SCAN	10 / 4096	Effect slots per insert / plugin parameters scanned for names.

11. Troubleshooting

Symptom	Fix
Force says <i>waiting for communication</i>	Check MIDI Settings: the DAW Control input uses the Akai Force script and is enabled, and the DAW Control output has the same Port number . Make sure Ableton isn't open and the Force is paired in the Akai Network Driver app.
Script isn't in the Controller type list	Click Update MIDI scripts in MIDI Settings or restart FL. Check that <code>Settings\Hardware\Akai Force Live</code> contains both .py files.
Script output: <i>Output port not linked</i>	Give the DAW Control output the same Port number as the input.
No <i>Akai Network</i> ports in FL	Install the Akai Network Driver, restart, pair the Force, and check both are on the same network.
PERFORM pads do nothing	Turn on Performance Mode in the playlist (the screen says <i>Turn on Perf Mode</i>).
Pads look white or pale	Those channels/tracks have FL's default grey. Give them colours (chapter 9).
A channel shows (<i>piano roll</i>) in STEPS	Its notes come from the piano roll. Play it in KEYS, or set <code>STEPS_HIDE_PIANO_ROLL = False</code> .
Hint bar: <i>FL refused that just now</i>	FL was busy (e.g. loading a project). Press again.
Brief disconnect while loading big projects	Normal. It reconnects by itself within a few seconds.
Two Forces/MPCs on the network	Only one can use Live Control with the computer at a time.
Preset name shows NEXT PRESET	That plugin doesn't tell FL its preset names (many VSTs don't). The preset still changes.
PLUGIN says <i>No plugin selected</i>	The selected channel is a plain sampler or empty. Select a channel with an instrument plugin, or switch to FX-chain mode (SHIFT + next device) for mixer effects.
Apple Silicon Mac	Not supported: Akai's network driver is Windows / Intel-Mac only.

12. For developers

File	Purpose
device_AkaiForce.py	The FL Studio controller script.
force_protocol.py	The Force's Live Control protocol: constants, message builders and the full control map. No dependencies.
force_plugin_maps.py	Your own knob pages per plugin (chapter 8).
fl_sim.py	Runs the script outside FL against a fake project. <code>python fl_sim.py</code> runs the automated checks (without FL installed: <code>pip install fl-studio-api-stubs</code>). <code>-live</code> drives a real Force. Needs <code>pip install mido python-rtmidi</code> .
force_probe.py	Talks to the Force with no DAW: handshake, message decoder, demo layout.
install.ps1 / install.bat / install.sh	Installers.
docs/build_manual.py	Builds this manual (HTML → PDF with Chrome or Edge).

Protocol summary

The protocol was worked out from Ableton Live's own *Akai_Force_MPC* remote script and verified on hardware. All SysEx is `F0 47 00 <product> <type> ... F7` (Force product id `0x40`).

Message	Format
Ping (host)	<code>F0 47 00 7F 00 F7</code> , about every 1-3 s
Pong (Force)	<code>F0 47 00 40 01 F7</code> . The Force also sends it on its own while waiting.
Text	<code>F0 47 00 40 10 id0 id1 lenHi lenLo ascii... F7</code> , e.g. track name <code>(0, t)</code> , clip name <code>(0, 16+row*8+col)</code> , knob screen <code>(18, 16+k)</code> , tempo <code>(3, 0)</code>
Pads	Channel 13 notes 16-79 (<code>16+row*8+col</code>); colour CC 40-103 on the same layout; the note velocity sets the clip state (0 empty, 2 stopped, 3 queued, 4 playing, 7 recording)
Pad pressure	Repeated note-ons while held: the first is the velocity, later ones are pressure, 0 = release
Knobs	Channel 14 CC 0-7 (mixer page) / 8-15 (device page), relative two's complement; touch = notes 0-15
Screen mixer	Channel 2-9 (one per strip): CC 0 volume, 1 pan, 3-6 sends, 124/125 meters; notes 0 solo, 1 mute, 5 arm
Colours	Values 8-77 = Live's 70-colour palette; 0-7 and 127 are state colours

The complete map is in `force_protocol.py`.

13. License and credits

Released under the **MIT License**. See `LICENSE` in the repository.

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Source, issues and updates: `github.com/Pyrodrifter/akai-force-fl-studio`